

Project Design Phase Document

Quotes Recommendation Bot using Rasa & NLP

This document represents the Project Design Phase of the Quotes Recommendation Bot. It defines system requirements, design scope, problem-solution fit, proposed solution, solution architecture, and data flow to ensure a scalable and intelligent chatbot system.

1. Project Objectives

- Design an intelligent chatbot for personalized quote recommendations.
- Utilize NLP for understanding user intent and context.
- Ensure scalability, performance, and modularity.
- Deliver an engaging and meaningful user experience.

2. Project Scope

In Scope:

- Text-based chatbot interface
- Mood and topic-based quote recommendations
- NLP-driven intent classification
- Backend recommendation logic

Out of Scope:

- Voice interaction
- Clinical mental health analysis

3. Functional Requirements

- Accept user input related to mood, emotion, or topic
- Identify intent and entities using NLP
- Retrieve and recommend relevant quotes
- Support feedback-based recommendations
- Maintain conversation context

4. Non-Functional Requirements

- Response time under 2 seconds
- High availability and fault tolerance
- Secure data handling
- Scalable architecture
- Easy system maintenance

5. Design Constraints

- Limited training dataset during initial phase
- Dependency on open-source frameworks
- Time and infrastructure constraints

6. Problem–Solution Fit

Problem:

Users often struggle to find meaningful and context-relevant quotes that match their mood or situation.

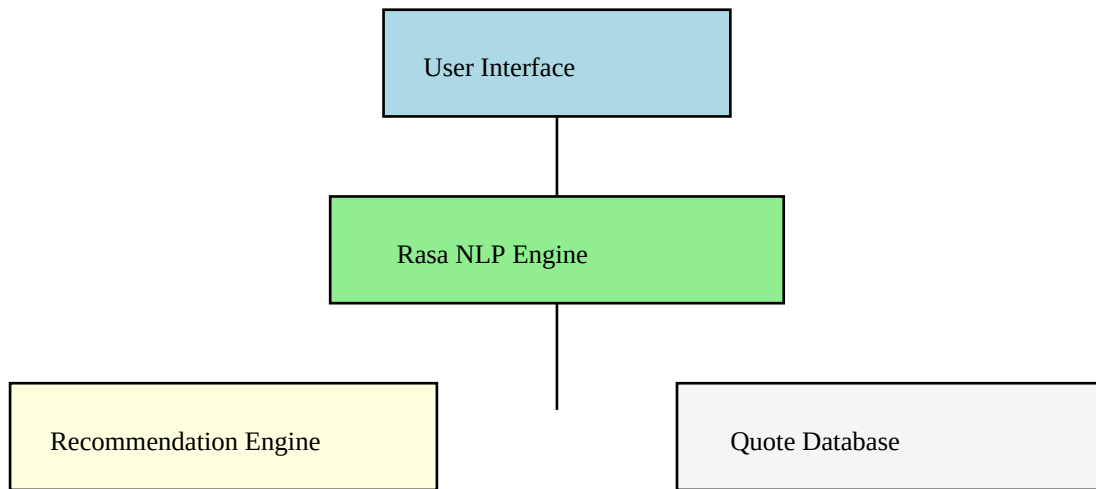
Solution:

An NLP-powered chatbot that understands user intent and emotional context to deliver personalized and relevant quotes instantly.

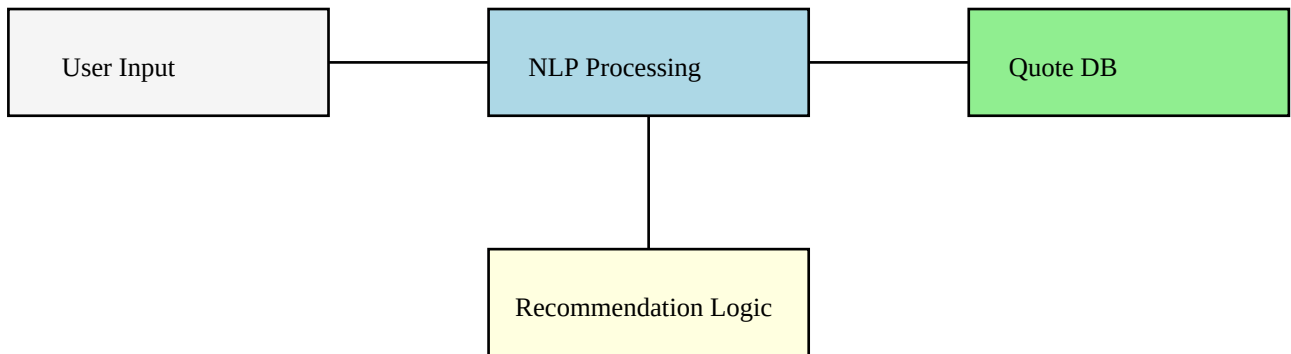
7. Proposed Solution

The proposed solution is a conversational chatbot built using Rasa and NLP techniques. The chatbot interacts with users, processes their input using NLP models, and recommends quotes using a rule-based and contextual recommendation engine.

8. Solution Architecture



9. Data Flow Diagram (DFD)



10. Technology Stack

Layer	Technology
Chatbot Framework	Rasa
NLP Processing	spaCy, TensorFlow
Backend	Python, FastAPI
Database	PostgreSQL / MongoDB
Deployment	Docker, AWS
Frontend	Web / Messaging Platforms

11. Conclusion

This Project Design Phase document defines a complete and professional design for the Quotes Recommendation Bot. The solution architecture and NLP-driven approach ensure intelligent recommendations, scalability, and an enhanced user experience.